

PERFORMANCE ANALYSIS OF VIRTUAL MACHINES AND CONTAINERS

EXPERIMENT 2 — MEMORY PERFORMANCE

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Experiment	Experiment 2 — Memory Performance
Benchmark	Sysbench Memory

Prepared using the supplied revised lab manual and the two Experiment 2 execution screenshots.

1. Objective

To measure memory-operation performance using Sysbench and compare the same workload between the Virtual Machine (VM) and Docker Container environments.

According to the lab manual, Experiment 2 uses Sysbench memory operations and records operations per second and latency as the main performance metrics. The same workload parameters are used in the VM and container so that the comparison is controlled. [filecite\turn1file0](#)

2. Experimental Parameters

Parameter	Lab-manual configuration
Tool	Sysbench 1.0.20
Workload	Memory operations
Memory block size	1M
Total memory size	10G
Threads	4
Operation shown in screenshots	Write
Metrics	Operations/sec and latency
Environments	VM and Docker Container

3. Procedure

The lab manual specifies the following VM command:

```
sysbench memory \  
--memory-block-size=1M \  
--memory-total-size=10G \  
--threads=4 \  
run
```

The manual then specifies repeating the VM benchmark 10 times and storing the outputs under results/raw/memory/vm. For Docker, the same Sysbench workload is executed through the vm-container-benchmark image and the results are stored under results/raw/memory/container. [filecite\turn1file0](#)

The manual also states that repeated tests are intended to provide statistically useful measurements and that the memory workload should be identical in both environments. [filecite\turn1file0](#)

4. VM Memory Benchmark — Execution Evidence

```

vm01@vm01-VMware-Virtual-Platform:~/vm-vs-container-performance$ mkdir -p ~/vm-vs-container-performance/results/raw/memory/vm
sysbench memory \
--memory.block-size=1M \
--memory.total-size=10G \
--threads=4 \
run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 4
Initializing random number generator from current time

Running memory speed test with the following options:
block size: 1024KiB
total size: 10240MiB
operation: write
scope: global

Initializing worker threads...

Threads started!

Total operations: 10240 (108900.87 per second)

10240.00 MiB transferred (108900.87 MiB/sec)

General statistics:
total time:                0.0931s
total number of events:    10240

Latency (ms):
min:                        0.02
avg:                        0.03
max:                        3.04
95th percentile:          0.03
sum:                        341.60

Threads fairness:
events (avg/stddev):       2560.0000/0.00
execution time (avg/stddev): 0.0854/0.01

```

Figure 1. VM memory benchmark: Sysbench 1.0.20, 4 threads, 1 MiB block size, 10 GiB total size, write operation.

Observed metric	VM result
Total operations	10,240
Operations/sec	108,900.87
Transferred	10,240.00 MiB
Transfer rate	108,900.87 MiB/sec
Total time	0.0931 s
Average latency	0.03 ms
Maximum latency	3.04 ms
95th percentile	0.03 ms
Latency sum	341.60 ms
Events/thread avg	2560.0000
Events/thread stddev	0.00
Execution time avg	0.0854 s
Execution time stddev	0.01 s

5. Docker Container Memory Benchmark — Execution Evidence

```

vn01@vn01-VMware-Virtual-Platform:~/vm-vs-container-performance$ mkdir -p ~/vm-vs-container-performance/results/raw/memory/container
docker run --rm \
vm-container-benchmark \
sysbench memory \
--memory-block-size=1M \
--memory-total-size=10G \
--threads=4 \
run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 4
Initializing random number generator from current time

Running memory speed test with the following options:
block size: 1024KiB
total size: 10240MiB
operation: write
scope: global

Initializing worker threads...

Threads started!

Total operations: 10240 (123725.89 per second)
10240.00 MiB transferred (123725.89 MiB/sec)

General statistics:
total time:                0.0819s
total number of events:    10240

Latency (ms):
min:                        0.02
avg:                        0.03
max:                        1.13
95th percentile:          0.03
sum:                        312.08

Threads fairness:
events (avg/stddev):       2560.0000/0.00
execution time (avg/stddev): 0.0780/0.00

```

Figure 2. Docker container memory benchmark: same Sysbench workload executed through vm-container-benchmark.

Observed metric	Container result
Total operations	10,240
Operations/sec	123,725.89
Transferred	10,240.00 MiB
Transfer rate	123,725.89 MiB/sec
Total time	0.0819 s
Average latency	0.03 ms
Maximum latency	1.13 ms
95th percentile	0.03 ms
Latency sum	312.08 ms
Events/thread avg	2560.0000
Events/thread stddev	0.00
Execution time avg	0.0780 s
Execution time stddev	0.00 s

6. VM vs Container Comparison

Metric	VM	Container
Operations/sec	108,900.87	123,725.89
Transfer rate	108,900.87 MiB/sec	123,725.89 MiB/sec

Total time	0.0931 s	0.0819 s
Average latency	0.03 ms	0.03 ms
Maximum latency	3.04 ms	1.13 ms
95th percentile	0.03 ms	0.03 ms
Latency sum	341.60 ms	312.08 ms

7. Observations

1. Both screenshots show the same benchmark structure: Sysbench 1.0.20, 4 threads, 1 MiB block size, 10 GiB total size, and a write operation.
2. The VM run reports 108,900.87 operations/sec, while the container run reports 123,725.89 operations/sec.
3. Both runs report an average latency of 0.03 ms.
4. The maximum latency visible in the VM screenshot is 3.04 ms; the container screenshot shows 1.13 ms.
5. The screenshots provide individual captured runs. They do not provide the complete set of ten repetitions required by the lab manual, so no ten-run statistical average is claimed here.

8. Result

The memory benchmark was successfully executed in both the VM and Docker environments using the controlled workload specified in Experiment 2. From the supplied execution evidence, the VM recorded 108,900.87 MiB/sec and the container recorded 123,725.89 MiB/sec.

9. Conclusion

Experiment 2 demonstrates how Sysbench can be used to measure memory-operation throughput and latency in a VM and a Docker container. For the two captured runs, the container screenshot records a higher transfer rate and a lower maximum latency. These are observations from the supplied runs; the lab manual calls for repeated measurements before drawing a statistically robust overall comparison.

10. Reference to Lab Manual

The relevant manual section is Experiment 2 — Memory Performance, followed by the resource-monitoring section. The manual specifies VM and Docker execution of the same Sysbench memory workload and recommends repeated runs.